

Admin User Manual

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1.Main page:

When you load in to the signed in main page, you will see a navigation column on the left-hand side of the page and pictures of the current lab in Manchester at the middle switching. (Figure 1)

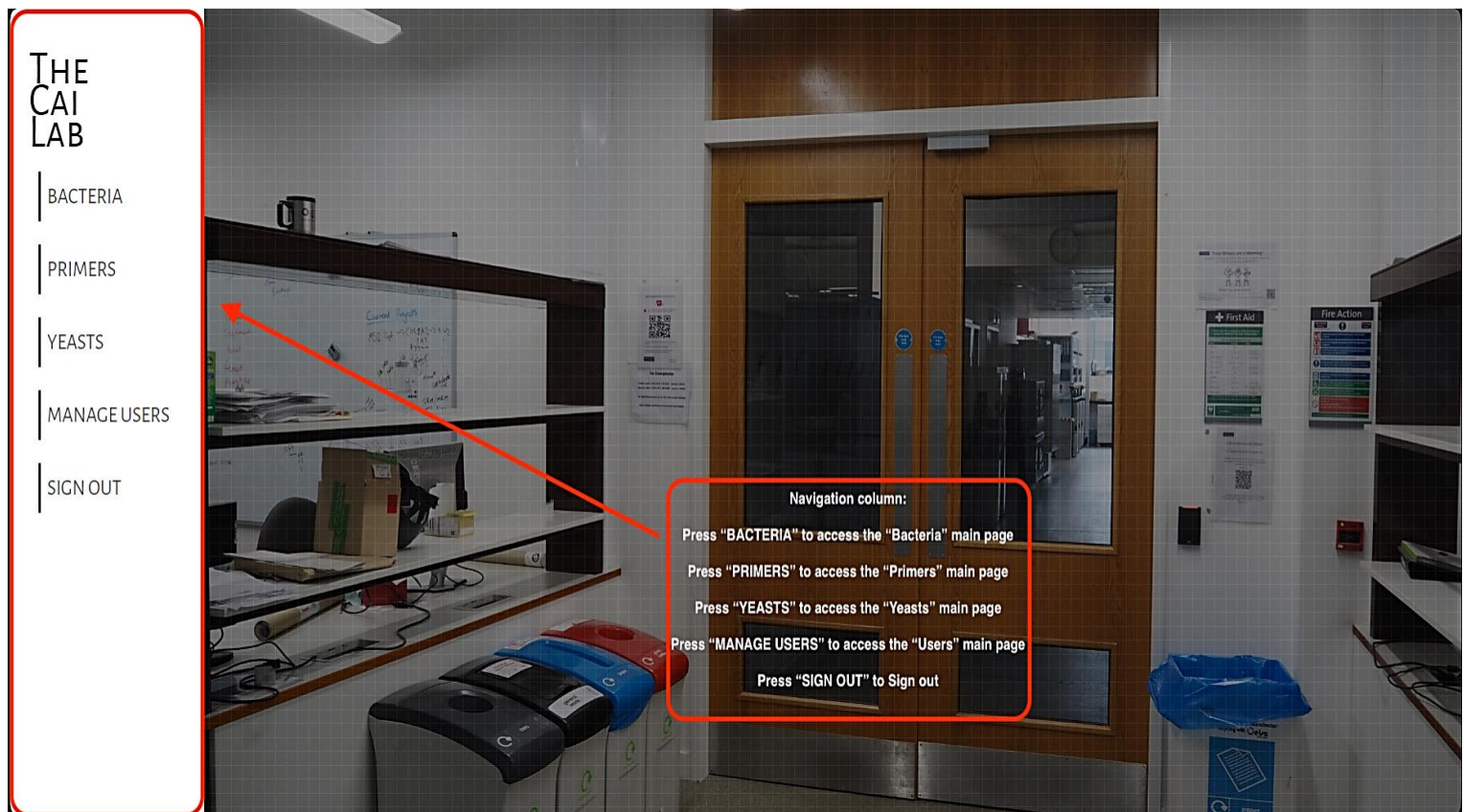


Figure 1: Signed in main page

2.Generic use for bacteria, yeast and primers:

The bacteria, primers, yeast pages follow the same structure. When you enter to either one of these pages, you will see a "select users" and "search all fields" filtering at the top left page under the title. You will see two buttons on the top right side, where it says, "Batch Create" and "Create New". Under all of it, you will see a table where it lists out the data that exists in the database. (Figure 2, 3, 4)

Bacteria

Select Users:

Justin Leung (19)

Search for all fields in Bacteria:

Search bacteria

Batch Create

Create New Bacterium

Show 10 entries

Personal Id	Lab Id	Plasmid Name	Alternative Name	Host Strain	Comments	Location	Date
JLe019	YCe8010	pSCW11-Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023-11-22 14:38:30
JLe018	YCe8009	pSCW11-Cre-EBD (URA3) test 16	pLM158, dae041	TOP10	FR18701117 from Jeff...	location B	2023-11-22 14:38:30
JLe017	YCe8008	pSCW11-Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023-11-22 11:14:46
JLe016	YCe8007	pSCW11-Cre-EBD (URA3) test 16	pLM158, dae041	TOP10	FR18701117 from Jeff...	location B	2023-11-22 11:14:46
JLe015	YCe8006	pSCW11-Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023-11-22 10:47:03
JLe014	YCe8005	pSCW11-Cre-EBD (URA3) test 16	pLM158, dae041	TOP10	FR18701117 from Jeff...	location B	2023-11-22

Figure 2: Bacteria Main page

Primers

Select Users:

Justin Leung (13)

Search for all fields in Primers:

Search primers

Batch Create

Create New Primer

Show 10 entries

Personal Id	Lab Id	Orientation	Melting Temperature	Concentration	Vendor	Location	Description
JLp013	YCp7188	Reverse	0				CDC27_40_target rv
JLp012	YCp7187	Forward	0				CDC27_40_target fw
JLp011	YCp7186	Reverse	12312312	12312312	12312312	12312312	12312312
JLp010	YCp7185	Forward	0	1	1	1	CDC27_40_target rv
JLp009	YCp7184	Forward	0	1	1	1	CDC27_40_target fw
JLp008	YCp7183	Forward	12312312	12312312	12312312	12312312	12312312
JLp007	YCp7182	Forward	0	1	1	1	CDC27_40_target rv
JLp006	YCp7181	Forward	0	1	1	1	CDC27_40_target fw
JLp005	YCp7180	Forward	0	1	1	1	CDC27_40_target rv

Figure 3: Primers Main page

Yeasts

Select Users:
Justin Leung (17) Search for all fields in Yeast:
Search yeast

Batch Create

Create New Yeast

Show 10 entries

Personal Id	Lab Id	Other Names	Mating Type	Plasmid	Genotype	Plasmid Type	Markers	Location	Comments	Parent Name	Other Parent Name	Date
JLy017	YCy5745	asdasd		asdasd	asdasd	asdasd	asdasd	asdasd	asdasd			2023-11-18 17:13:47.0
JLy016	YCy5744	BY4742	Alpha	pLeuOMeRS + pRS413-URA3_K209*	MAT Δ his3 Δ 1 leu2 Δ 0 lys2 Δ 0 ura3 Δ 0				Double Transformation...	JLy014	JLy014	2023-11-22 11:04:24.0
JLy015	YCy5743		Alpha	BY4742	MAT Δ his3 Δ 1 leu2 Δ 0 lys2 Δ 0 ura3 Δ 0		His3; Lys2; Ura3; Leu2		This is the yeast st...	JLy012	JLy014	2023-11-22 11:04:24.0
JLy014	YCy5742	BY4742		pLeuOMeRS + pRS413-URA3_K209*	MAT Δ his3 Δ 1 leu2 Δ 0 lys2 Δ 0 ura3 Δ 0				Double Transformation...			2023-11-22 10:57:20.0
JLy013	YCy5741		Alpha	BY4742	MAT Δ his3 Δ 1 leu2 Δ 0 lys2 Δ 0 ura3 Δ 0		His3; Lys2; Ura3; Leu2		This is the yeast st...	JLy012		2023-11-22 10:57:20.0
JLy012	YCy5740	BY4742		pLeuOMeRS	MAT Δ his3 Δ 1 leu2 Δ 0				Double			2023-11-

Figure 4: Yeast Main page

The following sections will give you detailed information regarding to the specific parts of the webpage. Note that most of the figures come from Bacteria pages.

2.1. Entries:

You can show how many entries to display with the select option located at the top left side above the table, you can choose 10, 25, 50 and 100 entries to display. The default choice is showing 10 entries per page. (Figure 5)

Bacteria

Select Users:
Justin Leung (19) Search for all fields in Bacteria:
Search bacteria

Batch Create

Create New Bacterium

Show 10 entries
10
25
50
100

Click on the select option between the text "Show" and "entries"

Choosing how many entries you want to display(10, 25, 50, 100)

Personal Id	Lab Id	Plasmid Name	Alternative Name	Host Strain	Comments	Location	Date
JLe019	YCe8010	pSCW11-Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023-11-22 14:38:30
JLe018	YCe8009	pSCW11-Cre-EBD (URA3) test 16	pLM158, dae041	TOP10	FR18701117 from Jeff...	location B	2023-11-22 14:38:30
JLe017	YCe8008	pSCW11-Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023-11-22 11:14:46
JLe016	YCe8007	pSCW11-Cre-EBD (URA3) test 16	pLM158, dae041	TOP10	FR18701117 from Jeff...	location B	2023-11-22 11:14:46

Figure 5: Entries function in Bacteria

2.2. Filtering:

There are two ways for filtering the table, you can select the user you want to see, and you can search all fields in the table. This is located between the show entries select option and title.

The following section will give you information how these two-filtering work.

2.2.1. Select user:

The options you can pick are all users and specific users, the number beside each specific users are the numbers of data that the specific user has created. The default selected user will be yourself, if you have created data within the specific table, if not the default selected user will be all users and it will not display your user within the selection box. (Figure 6)

THE
CAI
LAB

BACTERIA

PRIMERS

YEASTS

MANAGE USERS

SIGN OUT

Bacteria

Select Users:

Justin Leung (19)

Justin Leung (19)

All Users

Aimee H (25)

Alba Ballerini (296)

Alba Ubide (156)

Andrea Martella (555)

Claire Chung (1)

dabram a (44)

Daniel Schindler (309)

Elisa BM (4)

Ellie Payne (21)

Eloise O'Connor (129)

Eva Garcia-Ruiz (982)

Gabrielle David (5)

Hugo Villanueva (17)

James Sanders (110)

Jamie Auxillos (1779)

Joshua James (44)

Junyang Ong (2)

Katarina Jönsson (9)

Search for all fields in Bacteria:

Search bacteria

Click the select option below the "Select Users:" in the filtering section to display which user you want to display

Can pick yourself (If you have created data for this specific table), all users and other users

Batch Create

Create New Bacterium

Accession	Alternative Name	Host	Strain	Comments	Location	Date
CW11-11-22	pLM15, dae042	TOP10	FR18701117	location B	2023-11-22 14:38:30	
CW11-11-22	pLM158, dae041	TOP10	FR18701117	location B	2023-11-22 14:38:30	
CW11-11-22	pLM15, dae042	TOP10	FR18701117	location B	2023-11-22 11:14:46	

Figure 6: Select user in Bacteria

2.2.2. Search for all fields:

This field allows you to search anything from the table, it will search all data from the columns. You do not have to press the return/enter button to use this filter, the table will show words that you have typed in the search box automatically like a hot reload every time you have typed in a word. (Figure 7)

Bacteria

Select Users:

Justin Leung (19)

Search for all fields in Bacteria:

Search bacteria

Show 10 entries

Personal Id	Lab Id	Plasmid Name	Alternative Name	Host Strain	Comments	Location	Date
JLe019	YCe8010	pSCW11-Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023-11-22 14:38:30
JLe018	YCe8009	pSCW11-Cre-EBD (URA3) test 16	pLM158, dae041	TOP10	FR18701117 from Jeff...	location B	2023-11-22 14:38:30
JLe017	YCe8008	pSCW11-Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023-11-22 11:14:46

Just fill in the "Search for all fields in Bacteria" to search all columns in the table

No need to press return, it will auto-load every word you input

Batch Create

Create New Bacterium

Figure 7: Search all fields in Bacteria

2.3. Ordering:

The default order when you load the page are the dates, from the newest to the oldest. On each of the column title, you will see a grey diamond shaped button beside the title. (Figure 8) When you click on it the first time, it will order from the lowest to the highest or first alphabet to last alphabet from the specific column you have pressed from. (Figure 9) If you click it for the second time, the order of the specific column will be opposite. (Figure 10) However, you cannot do ordering with two columns at the same time.

Bacteria

Select Users:

Justin Leung (19)

Search for all fields in Bacteria:

Search bacteria

Show 10 entries

Personal Id	Lab Id	Plasmid Name	Alternative Name	Host Strain	Comments	Location	Date
JLe019	YCe8010	pSCW11-Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023-11-22 14:38:30
JLe018	YCe8009	pSCW11-Cre-EBD (URA3) test 16	pLM158, dae041	TOP10	FR18701117 from Jeff...	location B	2023-11-22 14:38:30
JLe017	YCe8008	pSCW11-Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023-11-22 11:14:46

Click the "diamond" shaped button to change the order of the table

Batch Create

Create New Bacterium

Figure 8: Ordering in Bacteria

Bacteria

Select Users: Search for all fields in Bacteria:

Show entries

Press the "diamond" button once to order the column from oldest to latest or first alphabet to last alphabet or first number to the last number

Personal Id	Lab Id	Plasmid Name	Alternative Name	Host Strain	Comments	Location	Date
ABe001	YCe6284	HcKan_P	HcKan_P	ccdB survival	HcKan_P vector used ...	Box1	2020-04-03 00:00:00
ABe002	YCe6285	HcKan_O	HcKan_O	ccdB survival	HcKan_O vector used ...	Box1	2020-04-03 00:00:00
ABe003	YCe6286	HcKan_T	HcKan_T	ccdB survival	HcKan_T vector used ...	Box1	2020-04-03 00:00:00
ABe004	YCe6287	POT1		ccdB survival	POT vector used in Y...	Box1	2020-04-03 00:00:00

Figure 9: Ordering in Bacteria when you click once

Bacteria

Select Users: Search for all fields in Bacteria:

Show entries

Press the "diamond" button twice to order the column from latest to oldest or last alphabet to first alphabet or last number to the first number

Personal Id	Lab Id	Plasmid Name	Alternative Name	Host Strain	Comments	Location	Date
YSe301	YCe5242	pRS415-YGR254W-YGR257C-YGR258C		DH5alpha	YGR254W, YGR257C and...		2018-03-19 00:00:00
YSe300	YCe4761	pCRCT-YGR139W		Beta-10	gRNA construct of YG...		2018-01-10 00:00:00
YSe299	YCe4760	pCRCT-YGR107W		Beta-10	gRNA construct of YG...		2018-01-10 00:00:00
YSe298	YCe4759	pCRCT-YGR076C		Beta-10	gRNA construct of YG...		2018-01-10 00:00:00

Figure 10: Ordering in Bacteria when you click twice

2.4. Selecting different pages:

At the bottom right of the table, you can access the next or previous page of the table if the table is not showing all the entries. It shows you the current page number and the next four-page number and the last page number. You can access the specific displayed page number by pressing on it. (Figure 11)

You can click the numbers to switch pages

Highlighted with blue box is your current page

You can click "Previous" or "Next" to also switch pages

		Cre-EBD (URA3) test 17			from Jeff...		11-22 10:47:03
JLe014	YCe8005	pSCW11-Cre-EBD (URA3) test 16	pLM158, dae041	TOP10	FR18701117 from Jeff...	location B	2023-11-22 10:47:03
JLe013	YCe8004	pSCW11-Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023-11-22 09:39:45
JLe012	YCe8003	pSCW11-Cre-EBD (URA3) test 16	pLM158, dae041	TOP10	FR18701117 from Jeff...	location B	2023-11-22 09:39:45
JLe011	YCe8002	aaa	aaaaaa	aaa	aaaa	aaa	2023-11-10 14:40:49
JLe010	YCe8001	pSCW11-Cre-EBD (URA3) test 16	pLM158, dae041	TOP10	FR18701117 from Jeff...	location B	2023-11-17 14:09:30

Showing 1 to 10 of 7,823 entries

[Previous](#) [1](#) [2](#) [3](#) [4](#) [5](#) ... [783](#) [Next](#)

Figure 11: Selecting different pages

2.5. Details of the data:

To access the detailed information of a specific “Personal Id” or “Lab Id” on the table, you can simply click either the “Personal Id” or “Lab Id” number which is coloured blue with a line under it. (Figure 12)

Inside the page, you will see the “Lab Id” and “Personal Id” at the top of the page. Below them, you will see the detailed data from the specific “Lab Id” and “Personal Id”. In addition, there is also two buttons located at the top right, it is the “New” (“Create”) and “Edit” button. (Figure 13)

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PRIMERS

YEASTS

MANAGE USERS

SIGN OUT

Bacteria

Select Users:
Justin Leung (19)

Search for all fields in Bacteria:
Search bacteria

Batch Create

Create New Bacterium

Show 10 entries

Personal Id	Lab Id	Plasmid Name	Alternative Name	Host Strain	Comments	Location	Date
JLe019	YCe8010	pSCW11-Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023-11-22 14:38:30
JLe018	YCe8009	pSCW11-Cre-EBD (URA3) test 16	pLM158, dae041	TOP10	FR18701117 from Jeff...	location B	2023-11-22 14:38:30
JLe017	YCe8008	pSCW11-Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023-11-22 11:14:46

Figure 12: How to access the details of a bacteria via “Personal Id” and “Lab Id”

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YEASTS
MANAGE USERS
SIGN OUT

JLe019, YCe8010

Lab Id	YCe8010
Personal Id	JLe019
Plasmid Name	pSCW11-Cre-EBD (URA3) test 17
Alternate Name	pLM15, dae042
Host Strain	TOP10
Comments	FR18701117 from Jeff lab Cre-EBD plasmid with URA3 marker
Location	location B
Marker	
Date	2023-11-22 14:38:30

Click “Edit” to edit this set of data from this page

Click “New” to create a new data for this specific table(In this picture, “Bacteria”)

New

Edit

Figure 13: Details of a bacteria page

2.6. Batch Create:

The “Batch Create” button is located at the top right of the webpage, it is beside the “Create New” button. (Figure 14) When you enter the “batch create” page, you will see a “choose file” button, a “Submit” button and “Download Batch Create Format” button. (Figure 15)

The “Download Batch Create Format” will give you a “csv” file with the structure to input the data for bacteria, primers or yeasts (Appendix 1, 2, 3). It is highly recommended for you to follow this structure and this file format for clean data in our database, or else the “batch create” will not accept your file and ask you to resubmit until you satisfy the requirements of the data requirements (Table 3.1, 3.2 , 3.3) and file format.

When you have finished inputting the data into the file, you may upload your file by clicking the “choose file” button, after that you click the “Submit” button to apply the batch create function. If your “batch create” file does not mee the data requirements (Table 3.1, 3.2, 3.3) or file requirements, it will pop up an error message stating which requirement you did not meet and ask you to resubmit it again until the requirements are met.

Bacteria

Select Users:

Justin Leung (19)

Search for all fields in Bacteria:

Search bacteria

Show 10 entries

Personal Id	Lab Id	Plasmid Name	Alternative Name	Host Strain	Comments	Location	Date
JLe019	YCe8010	pSCW11- Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023- 11-22 14:38:30
JLe018	YCe8009	pSCW11- Cre-EBD (URA3) test 16	pLM158, dae041	TOP10	FR18701117 from Jeff...	location B	2023- 11-22 14:38:30
JLe017	YCe8008	pSCW11- Cre-EBD (URA3) test 17	pLM15, dae042	TOP10	FR18701117 from Jeff...	location B	2023- 11-22 11:14:46

Click "Create New Bacterium" to access the "Create" page for "Bacteria"

Click "Batch Create" to access the "Batch Create" page

Batch Create

Create New Bacterium

Figure 14: How to access Batch create and Create from main Bacteria page

Batch Create Bacterium

To upload Batch create file to do batch create

Choose file No file chosen

4. Click the "Submit" button to upload the file, if it does not meet the data requirements, it will bounce you back to this page with a pop up message specifying which data requirement(s) you did not meet. If successful, it will return you to the main "bacteria" page for the "bacteria" as an example

Submit +

3. Upload your file by clicking the "Choose file" button, remember the file format must be "CSV"

To download the format

Download Bacteria Batch Create Format

1. Click "Download the Bacteria Batch Create Format" button to download the "Batch Create" format for the the specific table, for the "bacteria" as an example

AutoSave OFF bacteria-templat... Saved to this PC Search Yat Leung

File Home Insert Page Layout Formulas Data Review View Automate Help

Paste Clipboard Font Alignment Number Styles Cells Editing Analysis Sensitivity Add-ins

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Don't show again Save As...

A1

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
2	plasmid	alternative	host strain	bacterial other	location	comments													
3																			
4																			
5																			
6																			
7																			
8																			
9																			
10																			
11																			
12																			
13																			
14																			
15																			
16																			
17																			
18																			
19																			
20																			
21																			

bacteria-template

Ready Accessibility: Unavailable

Figure 15: Batch Create Page for Bacteria

2.7. Create/New:

The “create” bacteria, primers and yeasts button is located at the top right side of the page beside the “Batch Create” button (Figure 14), or it is located top right in the details page of your specific “personal Id” and “lab Id” beside the “Edit” button, but is it labelled as “New”. (Figure 13)

Within the “create” page, you will see a form to ask you to fill in based on the specific data requirements (Table 3.1, 3.2, 3.3). After you have filled in the form, you can press the “Submit” button at the bottom right of the “create” page to create the data. However, if the data that you have input does not meet the requirements for the specific data requirement, it will pop up an error message stating which requirement(s) you did not meet on the “create” page, and ask you to resubmit it again until you meet the requirements. (Figure 16)

The screenshot shows the 'Create Bacterium' form within a web application. On the left is a sidebar with the logo 'THE CAI LAB' and navigation links: BACTERIA, PRIMERS, YEASTS, MANAGE USERS, and SIGN OUT. The main form is titled 'Create Bacterium' and contains several input fields: 'Plasmid name *', 'Alternate name', 'Comments' (a large text area), 'Date *', 'Host Strain', 'Location', 'Bacterial marker', and 'Other bacterial marker'. A red box highlights the entire form. Below the form, a red arrow points to a red-bordered box containing the text '1. Fill in the details for Bacteria based on the bacteria data requirements'. A green arrow points from a green-bordered box containing '2. Press "Submit" to create bacteria' to a 'Submit +' button, which is also highlighted with a green border.

Figure 16: Create Page for Bacteria

2.8. Edit:

The “edit” bacteria, primers and yeasts button is located top right in the details page of your specific “Personal Id” and “Lab Id” beside the “New” (“Create”) button. (Figure 13)

Within the “edit” page, you will see a form to ask you to edit based on the specific data requirements (Table 3.1, 3.2, 3.3). After you have edited the form, you can press the “Submit” button at the bottom right of the “edit” page to edit the data. However, if the data that you have input does not meet the requirements for the specific data requirement, it will pop up an error message stating which requirement(s) you did not meet on the “edit” page and ask you to resubmit it again until you meet the requirements. (Figure 17)

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YEASTS
MANAGE USERS
SIGN OUT

Edit Bacterium

Plasmid name *

pSCW11-Cre-EBD (URA3) test 17

Choose file No file chosen

Alternate name

pLM15, dae042

Comments

FR18701117 from Jeff lab Cre-EBD plasmid with URA3 marker

Date *

2023-11-22 14:38

Host Strain

TOP10

Location

location B

Bacterial marker

Other bacterial marker

1. Make the changes

2. Click “Submit” to submit the changes

Submit +

Figure 17: Edit page for Bacteria

3. Data requirements:

Table 3.1 Bacteria:

Name(database)	Name(UI Create and Edit Page)	Type	Key	Null	Default	Comments
id	N/A	Int	Primary	No	None	Auto Increment
user_id	N/A	Int	Secondary	Yes	NULL	
plasmid_name	Plasmid Name	Varchar(255)	N/A	No	None	
alternate_name	Alternate Name	Varchar(255)	N/A	Yes	NULL	
host_strain	Host Strain	Varchar(255)	N/A	Yes	NULL	
comments	Comments	Text	N/A	Yes	NULL	
location	Location	Varchar(255)	N/A	Yes	NULL	
date	Date	Timestamp	N/A	No	NULL	
created_at	N/A	Timestamp	N/A	No	CURRENT_TIMESTAMP	Default Generated by database

updated_at	N/A	Timestamp	N/A	No	CURRENT_TIMESTAMP	Default Generated by database
bacterial_marker_id	Bacterial marker	Int	Secondary	Yes	NULL	-Selective Option on Create and Edit page -Linked to bacterial_markers table
other_bacterial_marker	Other bacterial marker	Varchar(255)	N/A	Yes	NULL	
lab_id	N/A	Varchar(255)	Secondary	No	None	Auto generated by code
personal_id	N/A	Varchar(255)	Secondary	No	None	Auto generated by code
edited	N/A	Tinyint(1)	N/A	No	0	
attachment_file_name	Choose file	Varchar(255)	N/A	Yes	NULL	File type in .DNA
attachment_content_type	N/A	Varchar(255)	N/A	Yes	NULL	
attachment_file_size	N/A	Int	N/A	Yes	NULL	
attachment_updated_at	N/A	Timestamp	N/A	Yes	NULL	
plasmid_data	N/A	Json	N/A	Yes	NULL	Not implemented
soft_delete	N/A	Tinyint(1)	N/A	No	0	Not implemented

Table 3.2 Primers:

Name(database)	Name(UI Create and Edit Page)	Type	Key	Null	Default	Comments
id	N/A	Int	Primary	No	None	Auto Increment
description	Description	Text	N/A	Yes	NULL	
sequence	Sequence	Text	N/A	Yes	NULL	
melting_temperature	Melting temperature	Int	N/A	Yes	NULL	
concentration	Concentration	Varchar(255)	N/A	Yes	NULL	
vendor	Vendor	Varchar(255)	N/A	Yes	NULL	
location	Location	Varchar(255)	N/A	Yes	NULL	
date	Date	Timestamp	N/A	No	CURRENT_TIMESTAMP	Default Generated, but must select a date on the edit and create page
comments	Comments	Text	N/A	Yes	NULL	
user_id	N/A	Int	Secondary	Yes	NULL	

created_at	N/A	Timestamp	N/A	No	CURRENT_TIMESTAMP	Default Generated
updated_at	N/A	Timestamp	N/A	No	CURRENT_TIMESTAMP	Default Generated
lab_id	N/A	Varchar(255)	Secondary	No	None	Auto generated by code
personal_id	N/A	Varchar(255)	Secondary	No	None	Auto generated by code
orientation_id	Orientation	Int	Secondary	Yes	NULL	-Selective Option on Create and Edit page -Linked to another orientations table
edited	N/A	Tinyint(1)	N/A	No	0	
plate_id	Plate ID	Int	N/A	Yes	NULL	
well_id	Well ID	VarChar(255)	N/A	Yes	NULL	
soft_delete	N/A	Tinyint(1)	N/A	No	0	Not implemented

Table 3.3 Yeast:

Name(database)	Name(UI Create and Edit Page)	Type	Key	Null	Default	Comments
id	N/A	Int	Primary	No	None	Auto Increment
user_id	N/A	Int	Secondary	Yes	NULL	
other_names	Other names	Varchar(255)	N/A	Yes	NULL	
plasmid	PlasmId	Varchar(255)	N/A	Yes	NULL	
genotype	Genotype	Varchar(255)	N/A	Yes	NULL	
plasmid_type	PlasmId type	Varchar(255)	N/A	Yes	NULL	
location	Location	Varchar(255)	N/A	Yes	NULL	
comments	Comments	Text	N/A	Yes	NULL	
date	Date	Timestamp	N/A	No	CURRENT_TIMESTAMP	Default generated
created_at	N/A	Timestamp	N/A	No	CURRENT_TIMESTAMP	Default generated
updated_at	N/A	Timestamp	N/A	No	CURRENT_TIMESTAMP	Default generated
other_mating_type	Other mating type	Varchar(255)	N/A	Yes	NULL	
mating_type_id	Mating type	Int	Secondary	Yes	NULL	-Selective Option on

						Create and Edit page -Linked to another mating_types table
lab_id	N/A	Varchar(255)	Secondary	No	None	Auto generated by code
personal_id	N/A	Varchar(255)	Secondary	No	None	Auto generated by code
edited	N/A	Tinyint(1)	N/A	No	0	
parent_name1	Parent Yeast Id	Varchar(255)	N/A	Yes	NULL	Parent name must exist within table, or else it will reject request for creating or editing
parent_name2	Other Parent Yeast Id	Varchar(255)	N/A	Yes	NULL	Parent name must exist within table, or else it will reject request for creating or editing
markers_list	Yeast Markers	Varchar(255)	N/A	Yes	NULL	

Table 3.4 Users:

Name(database)	Name(UI Create and Edit Page)	Type	Key	Null	Default	Comments
id	N/A	Int	Primary	No	None	Auto increment
deactivate	Deactivate	Tinyint(1)	N/A	No	0	Tick box selection
initials	N/A	Varchar(255)	Secondary	No	NULL	Auto generated by code
firstname	First Name	Varchar(255)	N/A	No	NULL	
lastname	Last Name	Varchar(255)	N/A	No	NULL	
email	Email	Varchar(255)	Secondary	No	None	
encrypted_password	- Encrypted password	Varchar(255)	N/A	No	None	Input the desired password, then the code will

	-New Password					generate an encrypted password for security purpose
reset_password_token	N/A	Varchar(255)	N/A	Yes	NULL	Auto generated by code from clicking the reset password
reset_password_sent_at	N/A	Timestamp	N/A	Yes	NULL	
sign_in_count	N/A	Int	N/A	No	0	
current_sign_in_at	N/A	Timestamp	N/A	No	CURRENT_TIMESTAMP	Default generated
last_sign_in_at	N/A	Timestamp	N/A	No	CURRENT_TIMESTAMP	Default generated
current_sign_in_ip	N/A	Varchar(255)	N/A	No	None	
last_sign_in_ip	N/A	Varchar(255)	N/A	No	None	
create_at	N/A	Timestamp	N/A	No	CURRENT_TIMESTAMP	Default generated
updated_at	N/A	Timestamp	N/A	No	CURRENT_TIMESTAMP	Default generated
admin	N/A	Tinyint(1)	N/A	No	0	
can_edit	N/A	Tinyint(1)	N/A	No	0	
approved	N/A	Tinyint(1)	N/A	No	0	
name	N/A	Varchar(255)	N/A	No	None	Auto generated by code
role_id	Role	Int	Secondary	No	1	-Selective Option on Create and Edit page -Linked to another user_role table

4. Manage users page:

When you load the page, you will see a search filter at the top of the page below the “Users” title. Beside the search filter you will see a “Create New Users” button. Below all of that, you will see the table listing all the users from the User table in the database. Each user has a “Reset Button” and “View” button located at the end of the right hand side of each row.

4.1. Search filter:

This field allows you to search anything from the table, it will search all data from the columns. You do not have to press the return/enter button to use this filter, the table will show words that you have typed in the search box automatically like a hot reload every time you have typed in a word.(Figure 18)

Users

Search for all fields in Users:

Create New Users

Name	Email	Level	Deactivate	Bacteria	Primers	Yeast	Reset Password	View
Yisha Luo	yishaluo@gmail.com	Test role	☐	0	0	0	<button>Reset Password</button>	<button>View</button>
Olivia Liu	no.1.liuwei@gmail.com	Test role	☐	1058	852	275	<button>Reset Password</button>	<button>View</button>
Roy Walker	100.days.rw@gmail.com	Test role	☐	273	1355	328	<button>Reset Password</button>	<button>View</button>
Yizhi Cai	caiyizhi@gmail.com	Test role	☐	220	455	662	<button>Reset Password</button>	<button>View</button>
Paulina Kanigowska	kanigowska.paulina@gmail.com	Test role	☐	72	115	53	<button>Reset Password</button>	<button>View</button>
Alba Ubide	albaubide@gmail.com	Test role	☐	156	210	99	<button>Reset Password</button>	<button>View</button>
Hugo Villanueva	hugo.villanueva92@gmail.com	Test role	☐	17	22	16	<button>Reset Password</button>	<button>View</button>
Yue Shen	shenyue226@gmail.com	Test role	☐	301	392	181	<button>Reset Password</button>	<button>View</button>
qi feng	fengqi0430@gmail.com	Test role	☐	8	41	7	<button>Reset Password</button>	<button>View</button>

Figure 18: Search filter in users main page

4.2. Ordering:

On each of the column title, you will see a grey diamond shaped button beside the title. When you click on it the first time, it will order from the lowest to the highest or first alphabet to the last alphabet from the specific column you have pressed on. If you click the second time, the order will be opposite. However, you cannot do ordering with two columns at the same time.(Figure 19)

Users

Search for all fields in Users:

Create New Users

Name	Email	Level	Deactivate	Bacteria	Primers	Yeast	Reset Password	View
Yisha Luo	yishaluo@gmail.com	Test role	☐	0	0	0	<button>Reset Password</button>	<button>View</button>
Olivia Liu	no.1.liuwei@gmail.com	Test role	☐	1058	852	275	<button>Reset Password</button>	<button>View</button>
Roy Walker	100.days.rw@gmail.com	Test role	☐	273	1355	328	<button>Reset Password</button>	<button>View</button>
Yizhi Cai	caiyizhi@gmail.com	Test role	☐	220	455	662	<button>Reset Password</button>	<button>View</button>
Paulina Kanigowska	kanigowska.paulina@gmail.com	Test role	☐	72	115	53	<button>Reset Password</button>	<button>View</button>
Alba Ubide	albaubide@gmail.com	Test role	☐	156	210	99	<button>Reset Password</button>	<button>View</button>
Hugo Villanueva	hugo.villanueva92@gmail.com	Test role	☐	17	22	16	<button>Reset Password</button>	<button>View</button>
Yue Shen	shenyue226@gmail.com	Test role	☐	301	392	181	<button>Reset Password</button>	<button>View</button>
qi feng	fengqi0430@gmail.com	Test role	☐	8	41	7	<button>Reset Password</button>	<button>View</button>

Figure 19: Ordering in users main page

4.3. Reset password:

If the specific user wants to reset the password for appropriate reasons, they must request the admin(s) to reset the password.

To reset the password, the admin will press the reset password button, it will pop up a reset password token/link for you to send to the specific user that wants to reset the password. The user will have 1 hour to reset the password, after this time, the token will expire, and the admin will need to generate another reset password token/link by clicking the “Reset Password” button if the user wishes to reset the password again.

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SIGN OUT

Users

Search for all fields in Users:

Create New Users

Name	Email	Level	Deactivate	Bacteria	Primers	Yeast	Reset Password	View
Yisha Luo	yishaluo@gmail.com	Test role	☐	0	0	0	Reset Password	View
Olivia Liu	no.1.liuwei@gmail.com	Test role	☐	1058	852	275	Reset Password	View
Roy Walker	100.days.rw@gmail.com	Test role	☐	273	1355	328	Reset Password	View
Yizhi Cai	caiyizhi@gmail.com	Test role	☐	220	455	662	Reset Password	View
Paulina Kanigowska	kanigowska.paulina@gmail.com	Test role	☐	72	115	53	Reset Password	View
Alba Ubide	albaubide@gmail.com	Test role	☐	156	210	99	Reset Password	View
Hugo Villanueva	hugo.villanueva92@gmail.com	Test role	☐	17	22	16	Reset Password	View
Yue Shen	shenyue226@gmail.com	Test role	☐	301	392	181	Reset Password	View
qi feng	fengqi0430@gmail.com	Test role	☐	8	41	7	Reset Password	View

Figure 20: Reset password button in user main page

4.4. Details of the user data:

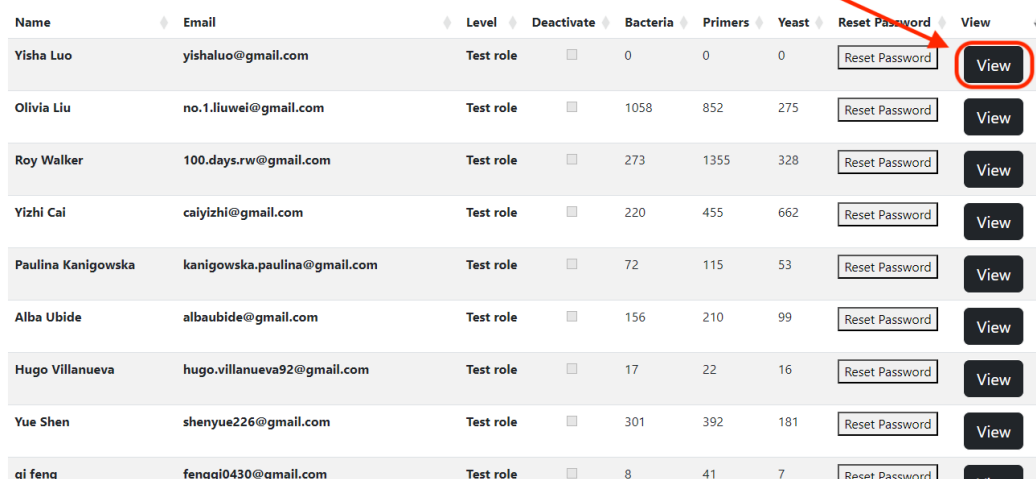
To access the detailed information of a specific user on the table, you can simply click the “View” button at the end of the right-hand side beside the “Reset Password” button. (Figure 21)

Users

Search for all fields in Users:

Search user

Create New Users



Name	Email	Level	Deactivate	Bacteria	Primers	Yeast	Reset Password	View
Yisha Luo	yishaluo@gmail.com	Test role	☐	0	0	0	Reset Password	View
Olivia Liu	no.1.liuwei@gmail.com	Test role	☐	1058	852	275	Reset Password	View
Roy Walker	100.days.rw@gmail.com	Test role	☐	273	1355	328	Reset Password	View
Yizhi Cai	caiyizhi@gmail.com	Test role	☐	220	455	662	Reset Password	View
Paulina Kanigowska	kanigowska.paulina@gmail.com	Test role	☐	72	115	53	Reset Password	View
Alba Ubide	albaubide@gmail.com	Test role	☐	156	210	99	Reset Password	View
Hugo Villanueva	hugo.villanueva92@gmail.com	Test role	☐	17	22	16	Reset Password	View
Yue Shen	shenyue226@gmail.com	Test role	☐	301	392	181	Reset Password	View
qi feng	fengqi0430@gmail.com	Test role	☐	8	41	7	Reset Password	View

Figure 21: “View” button in the user main page

Inside this page, you will see the “Lab Id” and “Personal Id” at the top of the page. Below them, you will see the detailed information from that user. In addition, there is also an “Edited” button located at the top right side of the page.

Displayed data in the detailed user information page:

Name(database)	Name(UI detailed user page)
id	Id
deactivate	Deactivate
initials	Initials
firstname	First Name
lastname	Last Name
email	Email
encrypted_password	Encrypted password
reset_password_token	Reset password token
reset_password_sent_at	Reset password sent at
sign_in_count	Sign in count
current_sign_in_at	Current sign in at
last_sign_in_at	Last sign in at
current_sign_in_ip	Current sign in ip
last_sign_in_ip	Last sign in ip
create_at	Created at
updated_at	Updated at
admin	N/A
can_edit	N/A
approved	N/A
name	Name
role_id	Role

4.5. Create Users:

The “Create New Users” button is located at the top of the main “Manage Users” page beside the search filter.

The screenshot shows the 'Users' management interface. On the left is a sidebar with navigation links: BACTERIA, PRIMERS, YEASTS, MANAGE USERS (highlighted), and SIGN OUT. The main content area is titled 'Users' and includes a search bar with the placeholder 'Search for all fields in Users:' and a 'Create New Users' button. A red callout box with an arrow points to this button, containing the text: 'Click “Create New Users” to access the “Create” page for a new user'. Below the search bar is a table listing users with columns for Name, Email, Level, Deactivate, Bacteria, Primers, Yeast, Reset Password, and View. The table contains 10 rows of user data.

Name	Email	Level	Deactivate	Bacteria	Primers	Yeast	Reset Password	View
Yisha Luo	yishaluo@gmail.com	Test role	☐	0	0	0	<button>Reset Password</button>	<button>View</button>
Olivia Liu	no.1.liuwei@gmail.com	Test role	☐	1058	852	275	<button>Reset Password</button>	<button>View</button>
Roy Walker	100.days.rw@gmail.com	Test role	☐	273	1355	328	<button>Reset Password</button>	<button>View</button>
Yizhi Cai	caiyizhi@gmail.com	Test role	☐	220	455	662	<button>Reset Password</button>	<button>View</button>
Paulina Kanigowska	kanigowska.paulina@gmail.com	Test role	☐	72	115	53	<button>Reset Password</button>	<button>View</button>
Alba Ubide	albaubide@gmail.com	Test role	☐	156	210	99	<button>Reset Password</button>	<button>View</button>
Hugo Villanueva	hugo.villanueva92@gmail.com	Test role	☐	17	22	16	<button>Reset Password</button>	<button>View</button>
Yue Shen	shenyue226@gmail.com	Test role	☐	301	392	181	<button>Reset Password</button>	<button>View</button>
qi feng	fengqi0430@gmail.com	Test role	☐	8	41	7	<button>Reset Password</button>	<button>View</button>

Figure 22: “Create New Users” button in user main page

Within the “Create” page, you will see a form to ask you to fill in based on the user data requirements(Section 3.4), it will ask you to fill in the “First Name”, “Last Name”, “Email”, “Password” and select a “Role” if the user. After you have filled in the form, you can press the “Submit” button toe add the user. However, if the data you have inputted does not meet the

The screenshot shows the 'Create users' form. It has a title 'Create users' and five input fields: 'First Name *', 'Last Name *', 'Email *', 'Password *', and 'Role *'. The 'Role' field is a dropdown menu with 'Admin' selected. A red callout box with an arrow points to the form fields, containing the text: '1. Fill in the details of the user'. A green callout box with an arrow points to the 'Submit +' button, containing the text: '2. Click “Submit” button to create the user'.

1. Fill in the details of the user

2. Click “Submit” button to create the user

requirements for the user data requirement, it will pop up an error message stating which part of your form has made a mistake, and ask you to resubmit it again.(Figure 23)

Figure 23: Create Users page

There are two options to allow the user to create their own personalised password:

1. The idea to allow the user to create their own personalised password is after the creation of the user, the admin will give you a reset password token/link to the user, then sent it to the user to allow them to create their own personalised password.
2. We ask the user and the admin to fill in the “Password” box together

Note that, there is no requirements for the “Password”, however we highly suggest the user to create a safe and secure password.

4.6. Edit Users:

The “Edit” button is located at top right in the details page of the specific user.(Figure 24)

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YEASTS

MANAGE USERS

SIGN OUT

Click “Edit” to “edit” the details for this specific user

Edit

Id	1
Email	yishaluo@gmail.com
First name	
Last name	
Name	Yisha Luo
Initials	
Role	Test role
Encrypted password	\$2a\$10\$hZWVz1nKr6SUljoJ4sLYNzDQQRLaKnaGZ0uQAxt/sc/0chD0g2.u
Reset password token	
Reset password sent at	
Sign in count	171
Current sign in at	2022-05-04 13:01:23
Last sign in at	2022-02-16 11:11:48
Current sign in ip	90.241.175.165
Last sign in ip	90.241.157.200
Created at	2014-03-06 10:30:19
Updated at	2022-09-28 14:36:34
Deactivate	☐

Figure 24: “Edit” button in the detailed user page

Within the edit page, you will see a form to ask you to edit based on the user data requirements(Section 3.4), it will ask you to edit the “First Name”, “Last Name”, “Email”, “Encrypted password”(do not edit this box), “New Password”, select a “Role” of the user and a “Deactivate” tick box. After you have filled in the form, you can press the “Submit” button to submit the changes you have made to the user. However, if the data you have inputted does not meet the requirements for the user data requirement, it will pop up an error message stating which part of your form has made a mistake, and ask you to resubmit it again.

The “Deactivate” tick box will disable the user, which will result in that user to not be allowed to login back to the page. (Figure 25)

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Edit Users

1. Make the changes

First Name- *

Yisha

Last Name *

Luo

Email *

yishaluo@gmail.com

Encrypted password * (do not edit this part)

\$2a\$10\$hZWVz1nKr6SUIyoJ4sLYNezDQQLaKnaGZ0uQAxT/sc/0chD0g2.u

New Password
(Leave it blank if you don't want to change)

Role *

Test role

Deactivate

Disable Account: ☐

2. Click the “Submit” button to submit the changes

Submit +

Figure 25: Edit Users page

5.Login and Sign out:

5.1. Sign in:

The “SIGN-IN” button is located at the left navigation column below the “WEBSITE” button on the before signed in main page. It will ask you to input your “Username” with your registered email and your “Password” that you have registered with this webpage. To login, you must press the “Log in” button, if one of your fields are incorrect, it will ask you to refill the “Username” and “Password” to log in again.(Figure 26)

1. Fill in the “Username” and “Password”

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Username
Password

Log in

2. Press “Log in” to log in

Figure 26: Sign in page

5.2. Sign out:

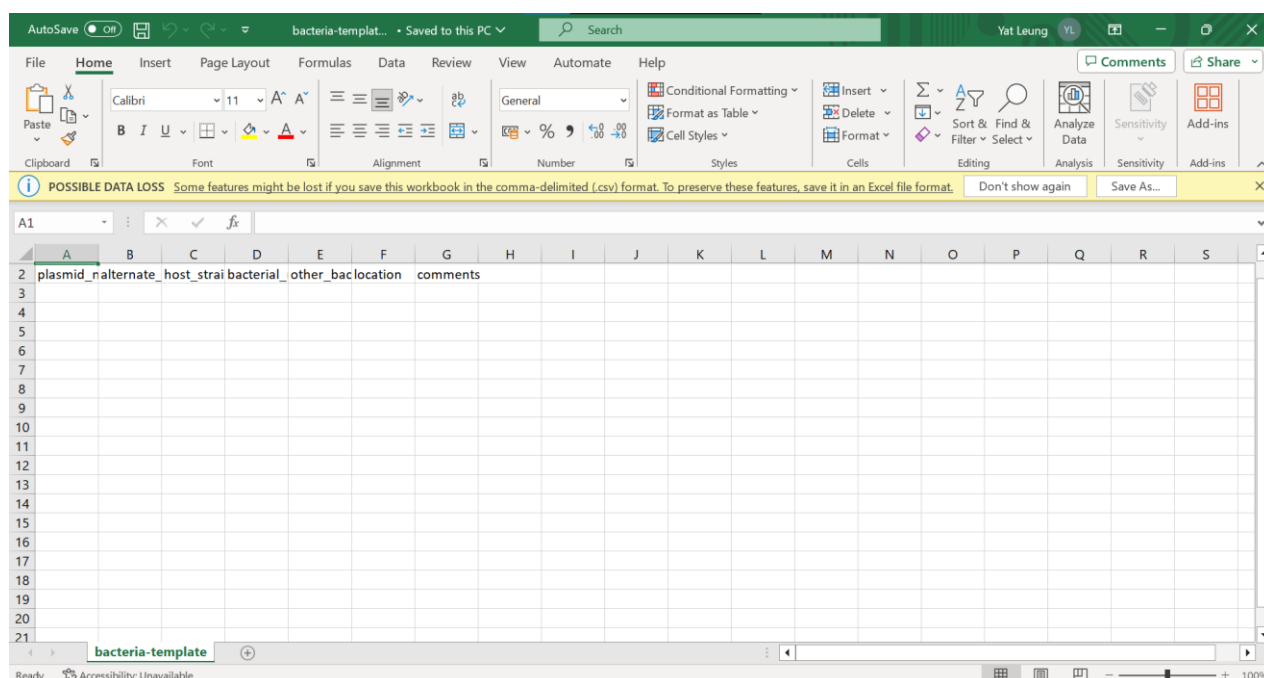
The “SIGN OUT” button is located at the left navigation column below “MANAGE USERS” button on the signed in pages. You simply need to click the “SIGN OUT” button to sign out.(Figure 1)

6.Security and accessibility:

Page name	Who can access it	Link
Login	Test role/Admin	http://straindb.cailab.org/login
Main page(Signed in)	Test role/Admin	http://straindb.cailab.org/indexSignIn
Bacteria**	Test role/Admin	http://straindb.cailab.org/bacteria
Primers**	Test role/Admin	http://straindb.cailab.org/primers
Yeasts**	Test role/Admin	http://straindb.cailab.org/yeast
Manage Users**	Admin	http://straindb.cailab.org/users
Note: ** = all pages under this directory		

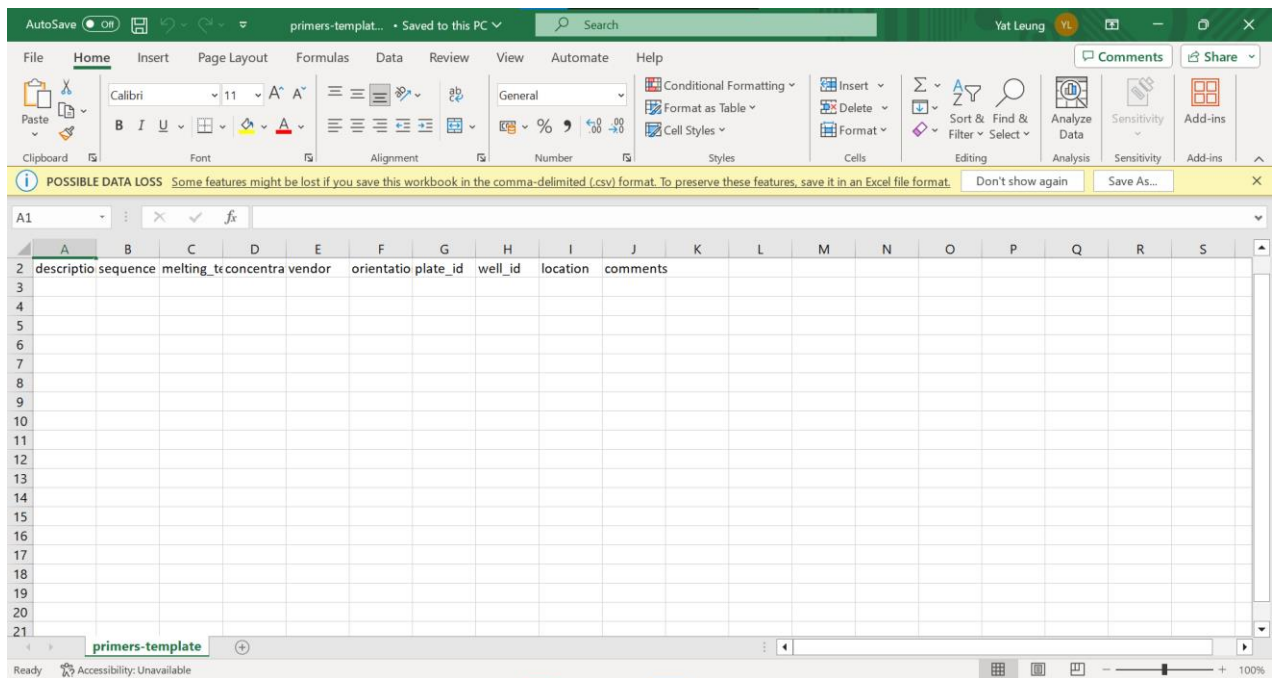
7.Appendix:

Appendix 1:



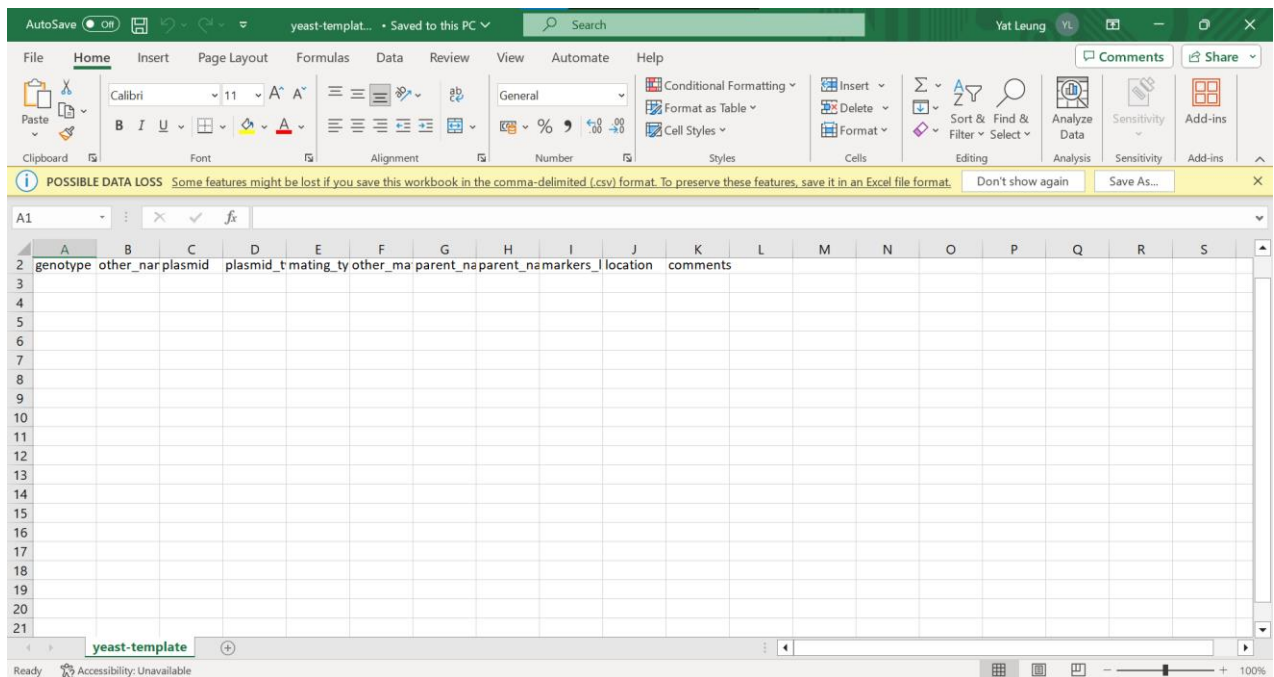
Appendix 1: Batch create structure for Bacteria

Appendix 2:



Appendix 2: Batch create structure for Primers

Appendix 3:



Appendix 3: Batch create structure for Yeast